Theory of Value

An Axiomatic Analysis of Economic Equilibrium
Gerard Debreu (1959)

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Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^K$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

Vectors in $\mathbb{R}^N$

a. Let $x^1 = \left(x_n^1\right)$ and $x^2 = \left(x_n^2\right)$ be two elements of $\mathbb{R}^N$; one defines their sum $x^1 + x^2$ as $\left(x_n^1 + x_n^2\right)$, i.e., the nth coordinate of $x^1 + x^2$ is the sum of the nth coordinates of x^1 and x^2. The element 0 has been defined (1.6.d) by the condition that all its coordinates are equal to 0. The negative of x is $-x = (-x_n)$. One writes $x^1 - x^2$ for $x^1 + \left(-x^2\right)$. If $x^1, \cdots, x^K, \cdots, x^K$ are K elements of $\mathbb{R}^N$, their sum is denoted $\sum_{n=1}^{K} x^K$.

Vectors in $\mathbb{R}^N$

b. Let $x = (x_n)$ be an element of $\mathbb{R}^N$, and t a real number; one defines their product tx, or xt, as (tx_n). Geometrically, given two points x^1 and x^2 of $\mathbb{R}^N$, one obtains $x^1 + x^2$ by completing the parallelogram having $0x^1$, $0x^2$ for sides; one obtains $-x$ by taking the symmetric of x with respect to 0: one obtains tx by representing $\mathbb{R}$ an the straight line $0x$, choosing 0 to represent 0, x to represent 1; then $t \in \mathbb{R}$ is represented by tx.

Vectors in $\mathbb{R}^N$

c. It is clear that the functions $\left(x^1, x^2\right) \to x^1 + x^2$ from $\mathbb{R}^N \times \mathbb{R}^N$ to $\mathbb{R}^N$, and $(t, x) \to tx$ from $\mathbb{R} \times \mathbb{R}^N$ to $\mathbb{R}^N$ are continuous.

Vectors in $\mathbb{R}^N$

d. The elements of $\mathbb{R}^N$ are also called vectors.

Vectors in $\mathbb{R}^N$

e. Given a vector a of $\mathbb{R}^N$, the transformation of $\mathbb{R}^N$ into itself defined by $x \to x + a$ is called the *a*-translation in $\mathbb{R}^N$.

Vectors in $\mathbb{R}^N$

f. Let X^1 and X^2 be two subsets of $\mathbb{R}^N$; one defines their sum $X^1 + X^2$ as the set of elements of $\mathbb{R}^N$ of the form $x^1 + x^2$ where $x^1 \in X^1$ and $x^2 \in X^2$. In other words, one takes, in all possible ways, an element of X^1 and an element of X^2 and adds them; the set of elements thus obtained is $X^1 + X^2$.

Vectors in $\mathbb{R}^N$

Example: (1) Let $X^1 = \left\{x \in \mathbb{R}^2 \mid 0 \leqq x_1 \leqq 1, x_2 = 0\right\}$, and $X^2 = \left\{x \in \mathbb{R}^2 \mid x_1 = 0, 0 \leqq x_2 \leqq 1\right\}$. Then $X^1 + X^2 = \left\{x \in \mathbb{R}^2 \mid 0 \leqq x_1 \leqq 1, 0 \leqq x_2 \leqq 1\right\}$. (2) See also Figure 4 in $\mathbb{R}^2$.

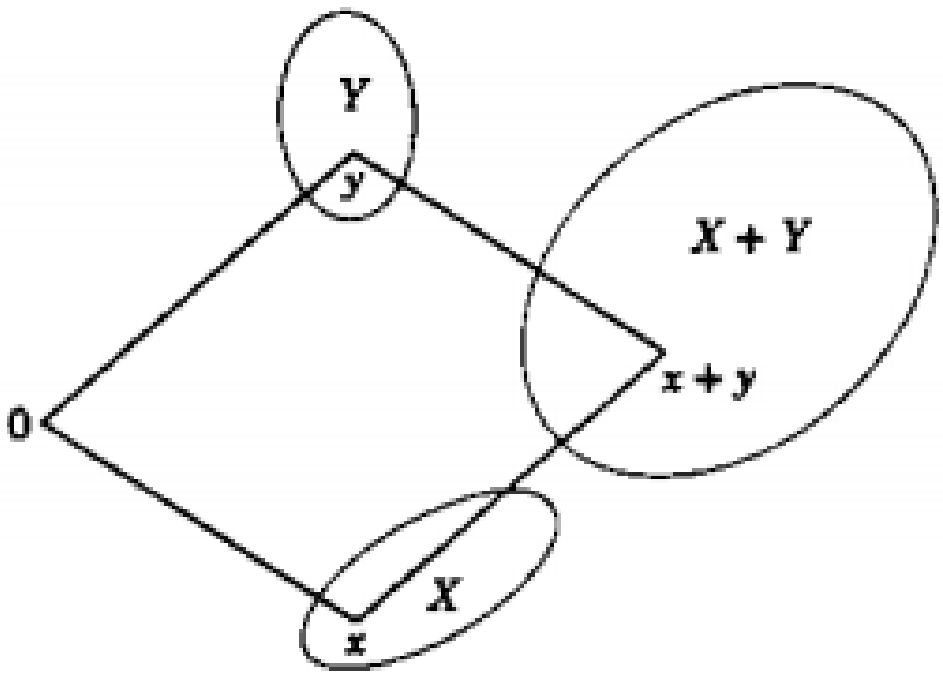


Fig. 4

Vectors in $\mathbb{R}^N$

One defines $-X$ as the set of elements of $\mathbb{R}^N$ of the form $-x$ where $x \in X$. One writes $X^1 - X^2$ for $X^1 + \left(-X^2\right)$. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, their sum is denoted $\sum_{k=1}^{K} X_k$. It is easy to prove :

(1) If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$.

That is, the sum of their adherences is contained in the adherence of their sum.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then for any $k = 1, \cdots, K$ and any $x_k \in \overline{X_k}$, there exists $\left\{x_k^q; q \in \mathbb{N}\right\} \subseteq X_k$ such that $\lim_{q\to\infty} x_k^q = x_k$. Then $\left\{\sum_{k=1}^{K} x_k^q; q \in \mathbb{N}\right\} \subseteq \sum_{k=1}^{K} X_k$ and $\lim_{q\to\infty} \sum_{k=1}^{K} x_k^q = \sum_{k=1}^{K} x_k \in \overline{\sum_{k=1}^{K} X_k}$, hence $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$. $\square$

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. In $\mathbb{R}^N$, a set is compact if and only if it is closed and bounded.

If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, then they are bounded, there exists a constant M_k such that $\|x_k\| \leqq M_k$ for $x_k \in X_k$, $k = 1, \cdots, K$. For the sum $\sum_{k=1}^{K} X_k$, any element x can be written as $x = \sum_{k=1}^{K} x_k$, where $x_k \in X_k$, thus

$$\|x\| = \left|\sum_{k=1}^{K} x_k\right| \leqq \sum_{k=1}^{K} |x_k| \leqq \sum_{k=1}^{K} M_k.$$

Therefore, the sum $\sum_{k=1}^{K} X_k$ is bounded.

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. The sum of closed sets is not necessarily closed. However, in $\mathbb{R}^N$, the sum of compact (which are closed and bounded) sets is closed. This is because the sum of compact sets can be viewed as the image of the compact set $\prod_{k=1}^{K} X_k$ under the continuous function $(x_1, \cdots, x_K) \mapsto \sum_{k=1}^{K} x_k$. Since the continuous image of a compact set is compact, the sum $\sum_{k=1}^{K} X_k$ is closed.

Since the sum of K compact subsets of $\mathbb{R}^N$ is both bounded and closed, it is compact.

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, for any sequence $\{x^q; q \in \mathbb{N}\} \subseteq \sum_{k=1}^{K} X_k$, we assume $x^q = \sum_{k=1}^{K} x_k^q$ such that $x_k^q \in X_k$ for all $k = 1, \cdots, K$. For compact set X_1, there exists a subsequence $\{x_1^{q_1}; q_1 \in \mathbb{N}\}$ of sequence $\{x_1^q; q \in \mathbb{N}\}$ such that $\lim_{q_1 \to \infty} x_1^{q_1} = x_1^0 \in X_1$. For compact set X_2, there exists a subsequence $\{x_2^{q_2}; q_2 \in \mathbb{N}\}$ of sequence $\{x_2^{q_1}; q_1 \in \mathbb{N}\}$ such that $\lim_{q_2 \to \infty} x_2^{q_2} = x_2^0 \in X_2$ with $\lim_{q_2 \to \infty} x_1^{q_2} = x_1^0 \in X_1$. Continuing this procedure, we have that, for compact set X_K, there exists a subsequence $\{x_K^{q_K}; q_K \in \mathbb{N}\}$ of sequence $\{x_K^{q_{K-1}}; q_{K-1} \in \mathbb{N}\}$ such that $\lim_{q_K \to \infty} x_K^{q_K} = x_K^0 \in X_K$ with $\lim_{k_K \to \infty} x_k^{q_K} = x_k^0 \in X_k$ for $k = 1, \cdots, K-1$.

Vectors in $\mathbb{R}^N$

Therefore

$$\lim_{q_K \to \infty} x_k^{q_K} = x_k^0 \in X_k, \qquad \text{for} \quad k = 1, \cdots, K$$

hence

$$\lim_{q_K \to \infty} \sum_{k=1}^{K} x_k^{q_K} = \sum_{k=1}^{K} x_k^0 \in \sum_{k=1}^{K} X_k$$

that is, there exists a subsequence $\{x^{q_K}; q_K \in \mathbb{N}\}$ of sequence $\{x^q; q \in \mathbb{N}\}$ such that

$$\lim_{q_K \to \infty} x^{q_K} = \lim_{q_K \to \infty} \sum_{k=1}^{K} x_k^{q_K} = \sum_{k=1}^{K} x_k^0 = x^0 \in \sum_{k=1}^{K} X_k$$

Thus, $\sum_{k=1}^{K} X_k$ are compact. $\square$

Vectors in $\mathbb{R}^N$

g. Let S be a subset of $\mathbb{R}^N$, and for every $k = 1, \cdots, K$, let f_k be a function from S to $\mathbb{R}^L$. Define a function f from S to $\mathbb{R}^L$ by $f(x) = \sum_{k=1}^{K} f_k(x)$ for every x in S. It is immediate that:

(3) If every f_k is continuous at the point x of S, then f is continuous at x.

(3′) The sum of K continuous functions at the point x of S is continuous at x.

Vectors in $\mathbb{R}^N$

h. Similarly let S be a subset of $\mathbb{R}^N$ and for every $k = 1, \cdots, K$, let T_k be a subset of $\mathbb{R}^L$ and φ_k be a correspondence from S to T_k. Consider the sum $T = \sum_{k=1}^{K} T_k$, and define a correspondence φ from S to T by $\varphi(x) = \sum_{k=1}^{K} \varphi_k(x)$ for every x in S. If every T_k is compact, so is T by (2). In this case one can prove (for lower semicontinuity the proof is immediate):

(4) If every φ_k is upper semicontinuous (resp. lower semicontinuous) at the point $x \in S$, then φ is upper semicontinuous (resp. lower semicontinuous) at x.

(4′) The sum of K upper semicontinuous (resp. lower semicontinuous) correspondences at the point $x \in S$ is upper semicontinuous (resp. lower semicontinuous) at x.

Vectors in $\mathbb{R}^N$

Proof. If $x^q \to x^0$, $y^q \in \varphi(x^q)$ and $y^q \to y^0$, then $y^q = \sum_{k=1}^{K} y_k^q$ with $y_k^q \in \varphi_k(x^q)$ and $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^q \to y_k^0$. If every φ_k is upper semicontinuous at the point $x^0 \in S$, then $y_k^0 \in \varphi_k\left(x^0\right)$. Thus

$$y^0 = \sum_{k=1}^{K} y_k^0 \in \sum_{k=1}^{K} \varphi_k\left(x^0\right) = \varphi\left(x^0\right)$$

φ is upper semicontinuous at x^0.

If $x^q \to x^0$ and $y^0 \in \varphi\left(x^0\right)$, then $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^0 \in \varphi_k\left(x^0\right)$. If every φ_k is lower semicontinuous at the point $x^0 \in S$, then there is (y_k^q) such that $y_k^q \in \varphi_k(x^q)$ and $y_k^q \to y_k^0$. Thus

$$y^q = \sum_{k=1}^{K} y_k^q \in \sum_{k=1}^{K} \varphi_k(x^q) = \varphi(x^q), \qquad y^q = \sum_{k=1}^{K} y_k^q \to \sum_{k=1}^{K} y_k^0 = y^0$$

φ is lower semicontinuous at x^0. □

Vectors in $\mathbb{R}^N$

i. Let x^1, x^2 be two points of $\mathbb{R}^N$, t^1, t^2 two real numbers such that $t^1 + t^2 = 1$. The point $t^1x^1 + t^2x^2$ is called the weighted average of x^1 and x^2 with weights t^1 and t^2 (respectively).

Vectors in $\mathbb{R}^N$

j. Let x, y be two different points of $\mathbb{R}^N$.

The straight line x, y is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, z = (1-t)\,x + ty\right\}$.

The closed half-line x, y (the origin x is written first) is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 \leqq t, z = (1-t)\,x + ty\right\}$.

The open half-line x, y is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 < t, z = (1-t)\,x + ty\right\}$.

It may be more suggestive to rewrite the expression of z as $z = x + t\,(y - x)$, i.e., one multiplies the fixed vector $(y - x)$ by the variable number t; this gives the straight line (or the half-line) 0, $(y - x)$; then one adds the fixed vector x (which amounts to a translation; see fig. 5 drawn for the case of the half-line).

Vectors in $\mathbb{R}^N$

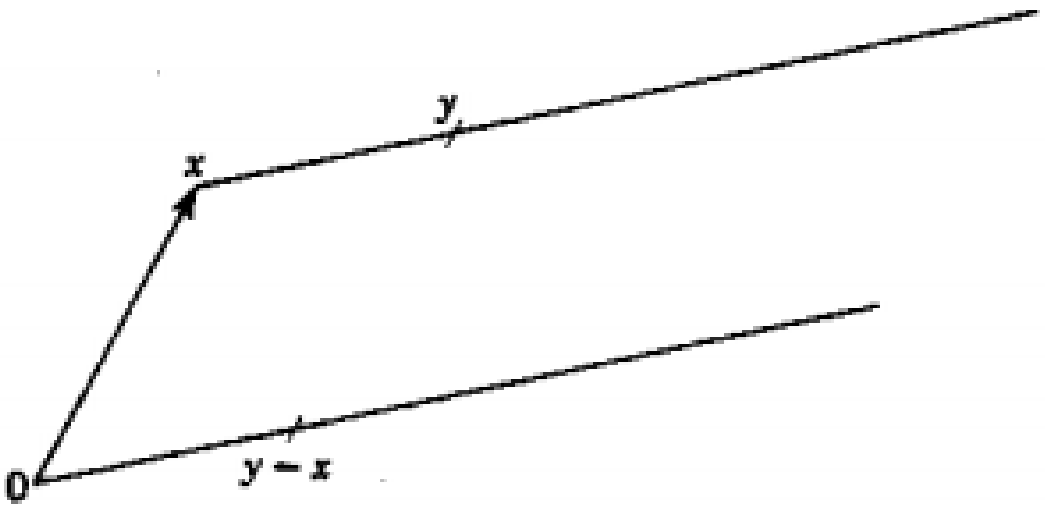


Fig. 5

Vectors in $\mathbb{R}^N$

k. Let x_1, x_2 be two points of $\mathbb{R}^N$ (different or not).
The closed segment x_1, x_2, denoted $[x_1, x_2]$, is
$\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 \leqq t \leqq 1, z = (1-t)\, x_1 + t x_2\right\}$. (See fig. 5.)
When $x_1 = x_2$, the closed segment $[x_1, x_2]$ is said to be degenerate.

Vectors in $\mathbb{R}^N$

I. Given a subset C of $\mathbb{R}^N$ and a point x in C, C is said to be a cone with vertex x if it contains the closed half-line x, y whenever it owns the point y (different from x).

"$y \in C$" implies "$x + t\,(y - x) \in C$ for $0 \leqq t$."

Vectors in $\mathbb{R}^N$

m. Given K cones $C_1, \cdots, C_k, \cdots, C_K$ with vertex 0, they are said to be positively semi-independent if "$x_k \in C_k$ for every k, and $\sum_{k=1}^{K} x_k = 0$" implies "$x_k = 0$ for every k," i.e., if it is impossible to take a vector in each cone so that their sum is 0, unless they are all equal to 0. It is clear that two cones C_1, C_2 with vertex 0 are positively semi-independent if and only if $C_1 \cap (-C_2) = \{0\}$.

Proof. If $C_1 \cap (-C_2) = \{0\}$ and for $x_1 \in C_1$ and $x_2 \in C_2$ with $x_1 + x_2 = 0$, then $x_1 = -x_2 \in C_1 \cap (-C_2) = \{0\}$, and hence $x_1 = x_2 = 0$. Thus, C_1 and C_2 are positively semi-independent.

If C_1 and C_2 with vertex 0 are positively semi-independent, and if $x \in C_1 \cap (-C_2)$, then $x \in C_1$ and $-x \in C_2$ with $x + (-x) = 0$, then $x = 0$. Hence $C_1 \cap (-C_2) = \{0\}$. $\square$

Vectors in $\mathbb{R}^N$

n. Consider a subset S of $\mathbb{R}^N$. To describe those of its points that are infinitely far from 0 one introduces the concept of its asymptotic cone, as follow. Let M be a non-negative real number, and denote by S^M the set $\{x \in S \mid \|x\| \geqq M\}$ of vectors in S whose norm is greater than or equal to M. Let $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing S^M (i.e., the intersection of all the closed cones with vertex 0 containing S^M). The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$, i.e.,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right);$$

it is clearly a closed cone with vertex 0.

Vectors in $\mathbb{R}^N$

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned} \Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \end{aligned}$$

Vectors in $\mathbb{R}^N$

Alfred Auslender and Marc Teboulle (2003) *Asymptotic Cones and Functions in Optimization and Variational Inequalities*

Definition 1.2. Let C be a non-empty set in $\mathbb{R}^N$. Then the asymptotic cone of the set C, denoted by C_∞, is the set of vectors $d \in \mathbb{R}^N$ that are limits in direction of the sequences $\{x_k; k \in \mathbb{N}\} \subseteq C$, namely

$$C_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

Vectors in $\mathbb{R}^N$

R. Tyrrell Rockafellar (1970) *Convex Analysis*

Definition 1.3. Let C be a non-empty convex set in $\mathbb{R}^N$.

- We shall say that C recedes in the direction D if C includes all the half-lines in the direction D which start at points of C. In other words, C recedes in the direction of y, where $y \neq 0$, if and only if $x + \lambda y \in C$ for every $\lambda \geqq 0$ and $x \in C$.
- The set of all vectors $y \in \mathbb{R}^N$ satisfying the latter condition, including $y = 0$, will be called the recession cone of C.
- The **recession cone** of C will be denoted by $\mathbf{0}^+ C$.

$$\mathbf{0}^+ C = \left\{ y \in \mathbb{R}^N \mid x + \lambda y \in C \text{ for all } x \in C \text{ and } \lambda \geqq 0 \right\}$$

- Directions in which C recedes will also be referred to as directions of recession of C.
- The recession cone of $\overline{C}$ has elsewhere been called the asymptotic cone of C.

$$\mathbf{A}C = \mathbf{0}^+ \overline{C}.$$

Vectors in $\mathbb{R}^N$

Proposition 2.1. Let $C \subseteq \mathbb{R}^N$ be non-empty.

(1) C_∞ is a closed cone.

(2) $\left(\overline{C}\right)_\infty = C_\infty$.

(3) If C is a cone, then $C_\infty = \overline{C}$.

Theorem 2.1. If S is a subset of $\mathbb{R}^N$, then $\mathbf{A}S = S_\infty$.

Proposition 2.2. Let $C \subseteq \mathbb{R}^N$ be non-empty and define the normalized set

$$C_N = \left\{ d \in \mathbb{R}^N \mid \text{there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \right.$$
$$\left. \text{with } \|x_k\| \to \infty \text{ such that } d = \lim_{k\to\infty} \frac{x_k}{\|x_k\|} \right\}$$

then $C_\infty = \text{Cone}\,(C_N)$, where for any set C,
$\text{Cone}\,(C_N) = \left\{\lambda x \in \mathbb{R}^N \mid x \in C_N, \lambda \geqq 0\right\}$.

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Vectors in $\mathbb{R}^N$

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \left\{d \in \mathbb{R}^N : \|d\| = 1\right\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! $\square$

Vectors in $\mathbb{R}^N$

o. It is obvious that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$".

Proof. If $S_1 \subseteq S_2$, then $S_1^M \subseteq S_2^M$ for $M \geqq 0$, thus $\Gamma\left(S_1^M\right) \subseteq \Gamma\left(S_2^M\right)$ for $M \geqq 0$, and hence $\mathbf{A}S_1 \subseteq \mathbf{A}S_2$.

One can prove that:

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}\left(S + \{x\}\right) = \mathbf{A}S$.

In other words, a translation of S does not alter its asymptotic cone. It is then easy to prove that:

(6) If $T \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}\left(S + T\right)$.

One can also prove that:

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof of (6). Let $x \in T$. Then $\mathbf{A}S = \mathbf{A}\left(S + \{x\}\right) \subseteq \mathbf{A}\left(S + T\right)$.

□

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S+\{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S+\{x\})$, we just need to let $S = (S+\{x\})+\{-x\}$ and we arrive at the conclusion. Let $T = S+\{x\}$.

For any $d \in \mathbf{A}(S+\{x\})$, $d \in (S+\{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S+\{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\|\frac{x'_k}{t_k} - d\right\| \leqq \left\|\frac{x'_k}{t_k} - \frac{x_k}{t_k}\right\| + \left\|\frac{x_k}{t_k} - d\right\| = \left\|\frac{x}{t_k}\right\| + \left\|\frac{x_k}{t_k} - d\right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$. $\square$

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

Vectors in $\mathbb{R}^N$

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. $\square$

Vectors in $\mathbb{R}^N$

p. It is now possible to give answers to the two important questions: When is the intersection of a collection of sets bounded? When is the sum of a finite number of closed sets closed? One can prove the intuitive result:

(8) Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

One can also prove:

(9) Given K closed subsets of $\mathbb{R}^N$, if their asymptotic cones are positively semi-independent, then their sum is closed.

This is a generalization of (2), since the asymptotic cone of a bounded set in $\{0\}$.

Vectors in $\mathbb{R}^N$

(8) Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. $\square$

Vectors in $\mathbb{R}^N$

(8) For all $\gamma \in \Gamma$, $S_\gamma \subseteq \mathbb{R}^N$, if $\bigcap\limits_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$, then $\bigcap\limits_{\gamma\in\Gamma} S_\gamma$ is bounded.

Proof. For all $\gamma \in \Gamma$, $\bigcap\limits_{\gamma\in\Gamma} S_\gamma \subseteq S_\gamma$, then $\mathbf{A}\left(\bigcap\limits_{\gamma\in\Gamma} S_\gamma\right) \subseteq \mathbf{A}(S_\gamma)$.

Whence $\mathbf{A}\left(\bigcap\limits_{\gamma\in\Gamma} S_\gamma\right) \subseteq \bigcap\limits_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap\limits_{\gamma\in\Gamma} S_\gamma\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap\limits_{\gamma\in\Gamma} S_\gamma$ is bounded.

□

Vectors in $\mathbb{R}^N$

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, a sufficient condition for the set $L(C)$ to be closed is that $L^{-1}\{0\} \cap C_\infty = \{0\}$. Under this condition one has $L(C_\infty) = L(C)_\infty$, although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, if $L^{-1}\{0\} \cap C_\infty = \{0\}$, then

(1) the set $L(C)$ is closed;

(2) $L(C_\infty) = L(C)_\infty$.

Although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Vectors in $\mathbb{R}^N$

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{N} \right\}$ is bounded, so we can extract a subsequence of $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{K} \right\}$ such that $\lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

Vectors in $\mathbb{R}^N$

Assuming henceforth that the condition holds, we prove now that $L(C)$ is a closed set. Suppose $y_k \to y$ with $y_k \in L(C)$, or in other words, $y_k = L(x_k)$ for some $x_k \in C$. The sequence $\{L(x_k)\}$ is bounded, so the sequence $\{x_k; k \in \mathbb{N}\}$ must be bounded as well, for the reasons just given, and it therefore has a cluster point x. We have $x \in C$ (because C is closed) and $y = L(x)$ (again because linear mappings are continuous), and therefore $y \in L(C)$.

Vectors in $\mathbb{R}^N$

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

Vectors in $\mathbb{R}^N$

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. $\square$

Vectors in $\mathbb{R}^N$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, a sufficient condition for $\sum_{n=1}^{N} C_n$ to be closed is the non-existence of vectors $x_n \in (C_n)_\infty$ such that $\sum_{n=1}^{N} x_n = 0$, except with $x_n = 0$ for all n. Then also

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty .$$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, if $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

(1) $\sum_{n=1}^{N} C_n$ is closed;

(2) $\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$.

Vectors in $\mathbb{R}^N$

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.3 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

Vectors in $\mathbb{R}^N$

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

Vectors in $\mathbb{R}^N$

(9) Given K closed subsets of $\mathbb{R}^N$, if their asymptotic cones are positively semi-independent, then their sum is closed.

Proof. Let $C = \prod_{n=1}^{N} C_n$. Then $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$ from Theorem 3.3. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n, \qquad \text{for} \quad x_n \in C_n,$$

then L is a linear mapping.

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

$$x_n \in (C_n)_\infty \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0 \quad \text{implies} \quad x_n = 0 \text{ for } n = 1, \cdots, N$$

$$(x_1, \cdots, x_N) \in \prod_{n=1}^{N} (C_n)_\infty \quad \text{and} \quad L((x_1, \cdots, x_N)) = 0 \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}.$$

For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, C is closed. By Theorem 3.5, we have $L(C) = \sum_{n=1}^{N} C_n$ is closed.

Vectors in $\mathbb{R}^N$

q. A subset of $\mathbb{R}^N$ is convex if it contains the closed segment $[x_1, x_2]$ whenever it contains the points x_1, x_2; in other words, if, whenever it contains the points x_1, x_2, it contains their weighted average with arbitrary positive weights. The set K in fig. 7.a, 7.b, o, 7.c is convex. The graph of fig. 2.a or 2 .b is not convex. The condition on a set that it be convex is crucial for economics.

Vectors in $\mathbb{R}^N$

r. A convex polyhedral cone is the sum of K closed half-lines.

Figure 6 pictures a convex polyhedral cone in $\mathbb{R}^2$ which is the sum of four, closed half-lines with origin 0. L_1 and L_3 are imagined to be in the plane of the page, L_2 in front of it, L_4 behind it.

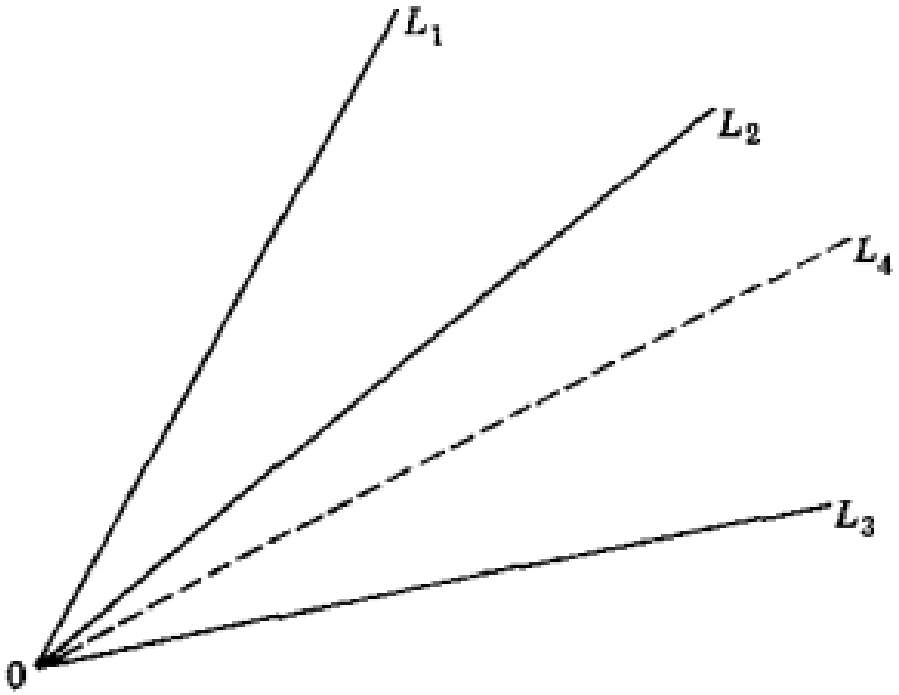


Fig. 6

Vectors in $\mathbb{R}^N$

s. It is easy to prove that:

(10) the intersection of a set of convex sets is convex,

(11) the sum and the product of K convex sets are convex,

(12) the adherence of a convex set is convex,

and also, using (8) of 1.6.u, that:

(13) a convex set is connected.

One can prove that:

(14) a closed, convex set owning 0, contains its asymptotic cone.

Vectors in $\mathbb{R}^N$

(10) the intersection of a set of convex sets is convex.

Proof of (10). If $\{X_\gamma; \gamma \in \Gamma\}$ is a family of convex subsets of X. Then for any x_1 and x_2 in $\bigcap_{\gamma\in\Gamma} X_\gamma$, then for any $\gamma \in \Gamma$, we have x_1 and x_2 in X_γ. For any $0 \leqslant t \leqslant 1$, we have $tx_1 + (1-t)\,x_2 \in X_\gamma$, hence $tx_1 + (1-t)\,x_2 \in \bigcap_{\gamma\in\Gamma} X_\gamma$. So $\bigcap_{\gamma\in\Gamma} X_\gamma$ is convex. $\square$

Vectors in $\mathbb{R}^N$

(11) the sum and the product of K convex sets are convex.

Proof of (11). Let $X_1, \cdots, X_K$ be convex. For any x^1 and x^2 in $\sum_{k=1}^{K} X_k$, there exist $x_k^1 \in X_k$ and $x_k^2 \in X_k$ for $k = 1, \cdots, K$ such that $x^1 = \sum_{k=1}^{K} x_k^1$ and $x^2 = \sum_{k=1}^{K} x_k^2$. For any $t \in [0, 1]$, we have

$$tx^1 + (1-t)\,x^2 = t\sum_{k=1}^{K} x_k^1 + (1-t)\sum_{k=1}^{K} x_k^2 = \sum_{k=1}^{K}\left[tx_k^1 + (1-t)\,x_k^2\right] \in \sum_{k=1}^{K} X_k$$

hence $\sum_{k=1}^{K} X_k$ is convex.

Let $x^i \in \prod_{k=1}^{K} X_k$ with $x^i = \left(x_1^i, \cdots, x_k^i, \cdots, x_K^i\right)$, $i =, 2$. For any $t \in [0, 1]$, we have

$$tx^1 + (1-t)\,x^2 = t\left(x_1^1, \cdots, x_k^1, \cdots, x_K^1\right) + (1-t)\left(x_1^2, \cdots, x_k^2, \cdots, x_K^2\right)$$

$$= \left(tx_1^1 + (1-t)x_1^2, \cdots, tx_k^1 + (1-t)x_k^2, \cdots, tx_K^1 + (1-t)x_K^2\right) \in \prod_{k=1}^{K} X_k$$

hence $\prod_{k=1}^{K} X_k$ is convex. $\square$

Vectors in $\mathbb{R}^N$

(12) the adherence of a convex set is convex.

Proof of (12). Let x^1 and x^2 in convex set $\overline{X}$. By the definition of closure, there exist sequences $\left\{x_q^1\right\} \subseteq X$ and $\left\{x_q^2\right\} \subseteq X$ with limits x^1 and x^2, respectively. Now, because X is convex, we know that for each x_q^1 and x_q^2, and for all $t \in [0,1]$, we have $tx_q^1 + (1-t)x_q^2 \in X$. This establishes the sequence $\left\{tx_q^1 + (1-t)x_q^2\right\} \subseteq X$ which is convergent by the convergence of $\left\{x_q^1\right\}$ and $\left\{x_q^2\right\}$. So we have that $tx^1 + (1-t)\,x^2 \in \overline{X}$, as $\overline{X}$ includes all limit points of X. This establishes that $\overline{X}$ is convex. □

Vectors in $\mathbb{R}^N$

(13) a convex set is connected.

Proof of (13). Suppose $E \subseteq \mathbb{R}^N$ is convex but not connected, then there exist non-empty open sets U_1 and U_2 in E such that $U_1 \cup U_2 = E$ and $U_1 \cap U_2 = \varnothing$. Let $x_1 \in U_1$ and $x_2 \in U_2$, We define

$$V_1 = \left\{ t \in [0,1] : tx^1 + (1-t)\, x^2 \in U_1 \right\}$$

and

$$V_2 = \left\{ t \in [0,1] : tx^1 + (1-t)\, x^2 \in U_2 \right\}$$

then $V_1 \neq \varnothing$ and $V_2 \neq \varnothing$ and $V_1 \cup V_2 = [0,1]$ and $V_1 \cap V_2 = \varnothing$. As U_1 and U_2 are open, hence V_1 and V_2 are also open on $[0,1]$. Hence $[0,1]$ is disconnected, contradicting (8) of 1.6.u. Hence E is connected. $\square$

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\frac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda \frac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. $\square$

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Assume $d \in C_\infty$: By definition of the asymptotic cone, there exists a sequence $\{x_k\} \subseteq C$ and a sequence $\{t_k\} \subseteq \mathbb{R}$ with $t_k \to \infty$, such that $\dfrac{x_k}{t_k} \to d$.

- Since C is convex and contains 0, for each k, the point $\lambda_k x_k$ lies in C for $\lambda_k \in [0,1]$. Choose $\lambda_k = \dfrac{1}{t_k}$, then $\lambda_k x_k = \dfrac{x_k}{t_k}$.
- As $t_k \to \infty$, $\lambda_k \to 0$, and $\dfrac{x_k}{t_k} \to d$.
- Since C is closed and $\dfrac{x_k}{t_k} \in C$ for all k, the limit d must also belong to C, $d \in C$.

Since $d \in C_\infty$ implies $d \in C$, we have $C_\infty \subseteq C$.

Vectors in $\mathbb{R}^N$

t. Let S he a subset of $\mathbb{R}^N$. Its convex hull, denoted $\dot{S}$, is defined as the intersection of all the convex sets containing S. According to (10), $\dot{S}$ is indeed convex and can therefore also be characterized as the smallest convex set containing S. One can prove that:

(15) Given K subsets of $\mathbb{R}^N$, the convex hull of their sum is equal to the sum of their convex hulls.

The closed convex hull of S is, by definition, the adherence $\overline{\dot{S}}$ of the convex hull of S. According to (3) of 1.6.j, $\overline{\dot{S}}$ is indeed closed; according to (12), it is indeed convex. It is easy to see that a closed convex set containing S necessarily contains $\overline{\dot{S}}$, which can therefore also he characterized as the smallest closed convex set containing S.

Vectors in $\mathbb{R}^N$

(15) Given K subsets of $\mathbb{R}^N$, the convex hull of their sum is equal to the sum of their convex hulls.

(15) If $S_k \subseteq \mathbb{R}^N$ for $k = 1, \cdots, K$, denote $S = \sum_{k=1}^{K} S_k$, then $\dot{S} = \sum_{k=1}^{K} \dot{S}_k$.

$$\text{conv}(A) = \left\{ \sum_{i=1}^{I} \lambda_i x_i \mid x_i \in A, \lambda_i \geqq 0, \sum_{i=1}^{I} \lambda_i = 1, I \in \mathbb{N} \right\}.$$

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

Let $x \in \dot{S}$. By definition, x is a convex combination of points in $S = \sum_{k=1}^{K} S_k$. $x = \sum_{i=1}^{I} \lambda_i \sum_{k=1}^{K} x_{ki}$, where $x_{ki} \in S_k$, $\lambda_i \geqq 0$, and $\sum_{i=1}^{I} \lambda_i = 1$. Rearranging the sum, we have $x = \sum_{k=1}^{K} \sum_{i=1}^{I} \lambda_i x_{ki}$. Each term $\sum_{i=1}^{I} \lambda_i x_{ki}$ is a convex combination of points in S_k, so $\sum_{i=1}^{I} \lambda_i x_{ki} \in \dot{S}_k$. Therefore $x \in \sum_{k=1}^{K} \dot{S}_k$. This proves $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

Let $x \in \sum_{k=1}^{K} \dot{S}_k$. By definition, $x = \sum_{k=1}^{K} x_k$, where $x_k \in \dot{S}_k$ for $k = 1, \cdots, K$. Each x_k is a convex combination of points in S_k. $x_k = \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$, where $x_{ki_k} \in S_k$, $\lambda_{ki_k} \geqq 0$, and $\sum_{i_k=1}^{I_k} \lambda_{ki_k} = 1$.

Substituting into x, we have $x = \sum_{k=1}^{K} \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$. This is a convex combination of points in $S = \sum_{k=1}^{K} S_k$, because $\sum_{k=1}^{K} \sum_{i=1}^{I_k} \lambda_{ki} = \sum_{k=1}^{K} 1 = K$ and we can normalize the coefficients to form a convex combination. Therefore $x \in \dot{S}$. This proves $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

Vectors in $\mathbb{R}^N$

$$
\begin{aligned}
x &= \sum_{k=1}^{K}\sum_{i_k=1}^{I_k}\lambda_{ki_k}x_{ki_k} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}x_{1i_1} + \cdots + \sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{1i_1} + \cdots + \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\left(x_{1i_1} + \cdots + x_{Ki_K}\right) = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\sum_{k=1}^{K}x_{ki_k} \\
&= \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\lambda_{1i_1}\cdots\lambda_{Ki_K}\right]\sum_{k=1}^{K}x_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\prod_{k=1}^{K}\lambda_{ki_k}\right]\left[\sum_{k=1}^{K}x_{ki_k}\right]
\end{aligned}
$$

where $\sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\prod_{k=1}^{K}\lambda_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{1i_1}\cdots\lambda_{Ki_K} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K} = 1$, then $x \in \dot{S}$. $\square$

Vectors in $\mathbb{R}^N$

u. Let $x^1 = \left(x_n^1\right)$ and $x^2 = \left(x_n^2\right)$ be two elements of $\mathbb{R}^N$; one defines their inner product $x^1 \cdot x^2$ as the number $\sum_{n=1}^{N} x_n^1 x_n^2$. It is clear that the function $\left(x^1, x^2\right) \to x^1 \cdot x^2$ from $\mathbb{R}^N \times \mathbb{R}^N$ to $\mathbb{R}$ is continuous.

When $x^1 \cdot x^2 = 0$. one says that x^1 and x^2 are orthogonal. In the case of $\mathbb{R}^2$, visualized as in 1.6.c. saying that two vectors x^1 and x^2 different from 0 are orthogonal is equivalent to saying that the half-lines $0, x^1$ and $0, x^2$ are perpendicular, provided that the units on the three axes have the same length.

Vectors in $\mathbb{R}^N$

v. Let p he an element of $\mathbb{R}^N$ different from 0, and c a real number; the set $H = \left\{ z \in \mathbb{R}^N \mid p \cdot z = c \right\}$ is a hyperplane with normal p. If z' is a point in H (one says also that H goes through z'), one has $p \cdot z' = c$ and the above expression may be rewritten $H = \left\{ z \in \mathbb{R}^N \mid p \cdot (z - z') = 0 \right\}$. Thus the hyperplane H is the set of points z of $\mathbb{R}^N$ such that $z - z'$ is orthogonal to p; p and H are said to be orthogonal. If p and c are multiplied by the same real number different from 0, the hyperplane H is unchanged. An intersection of hyperplanes is called a linear manifold.

Vectors in $\mathbb{R}^N$

w. Given a hyperplane H with normal p, the point z of $\mathbb{R}^N$ is said to be above H if $p \cdot z > c$. The closed half-space above H is $\left\{z \in \mathbb{R}^N \mid p \cdot z \geqq c\right\}$. One obtains similar definitions replacing above, $>, \geqq$ by below, $<, \leqq$. A closed half-space is easily seen to be closed and convex. So is a hyperplane, since it is the intersection of two closed half-spaces, and hence a linear manifold.

Vectors in $\mathbb{R}^N$

x. A hyperplane H is said to be bounding for a subset S of $\mathbb{R}^N$ if S is contained in one of the two closed half-spaces determined by H. In other words, H is bounding for S if S is entirely on one side of H with, possibly, points in it. One can prove the fundamental theorem:

(16) (Minkowski's Theorem) Let K be a convex subset of $\mathbb{R}^N$ and x a point of $\mathbb{R}^N$. There is a hyperplane H through x and bounding for K if and only if x is not interior to K.

Vectors in $\mathbb{R}^N$

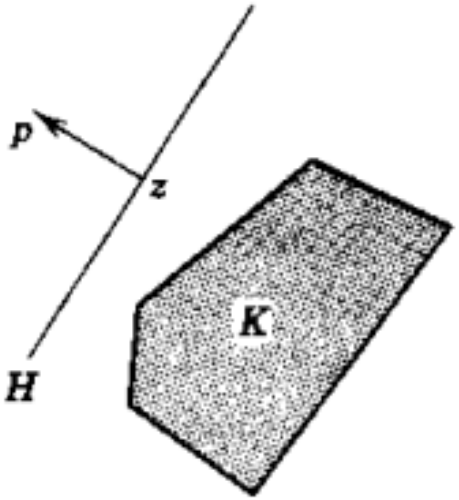


Fig. 7.a

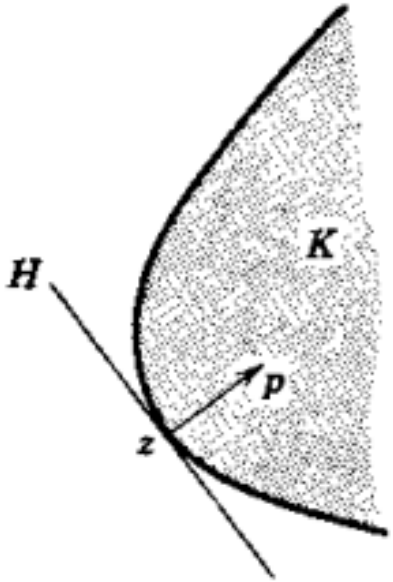


Fig. 7.b

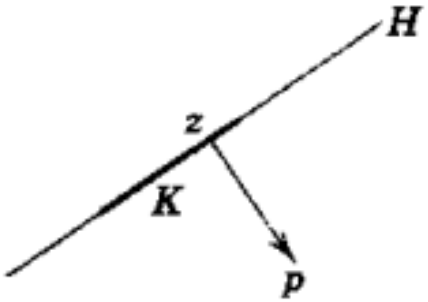


Fig. 7.c

The intuitive content of the result in R^2 is brought out by fig. 7.a, where z is exterior to K, by fig. 7.b, where z is in the boundary of K, and by fig. 7.c, where K (the heavy-lined segment) has no interior. It is easy to draw a non-convex set in R^2 for which some non-interior points are in no bounding hyperplane.

Vectors in $\mathbb{R}^N$

y. Let C be a cone with vertex 0. Its polar is the set $C^\circ = \{x \in \mathbb{R}^N \mid x \cdot x' \leqq 0 \text{ for every } x' \in C\}$. It is easy to see that C° is a closed, convex cone with vertex 0.

Figure 8 provides an illustration in $\mathbb{R}^2$.

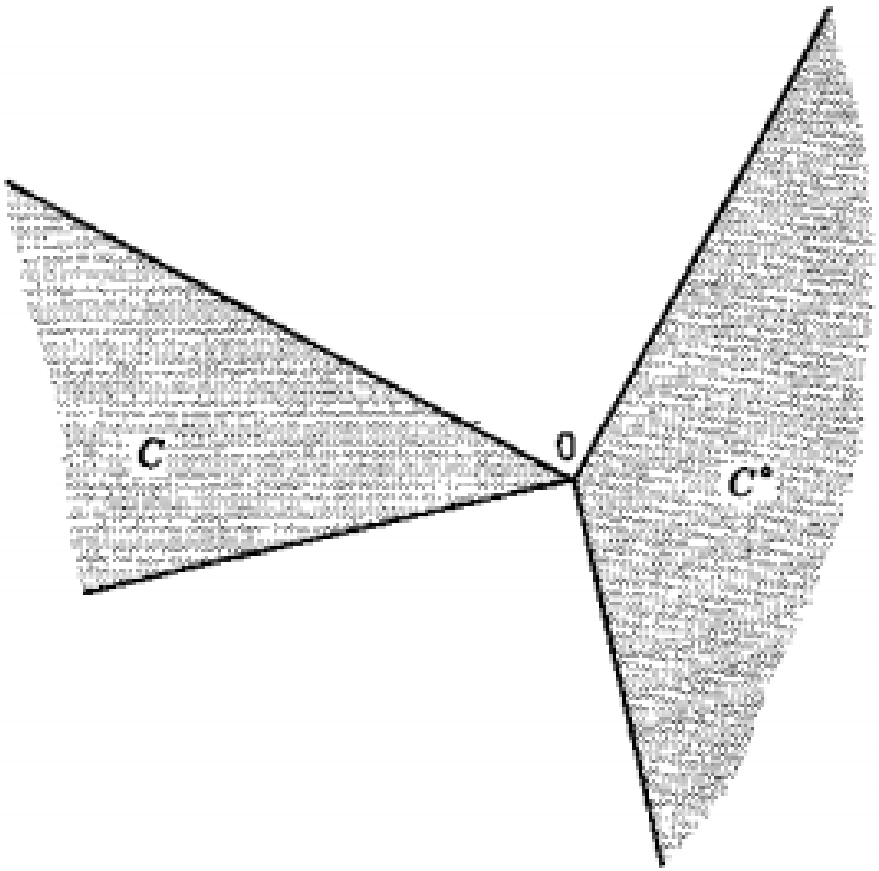


Fig. 8

Vectors in $\mathbb{R}^N$

z. According to the general convention of 1.4.j, an ordering is defined on $\mathbb{R}^N$ by $x^1 \leqq x^2$ if $x_n^1 \leqq x_n^2$ for every $n = 1, \cdots, N$. As remarked in 1.4.j, $x^1 < x^2$ means "$x_n^1 \leqq x_n^2$ for all n and $x_n^1 < x_n^2$ for at least one n," and $x^1 \ll x^2$ means "$x_n^1 < x_n^2$ for all n." (Several authors have used the notation $\leqq, \leq, <$ respectively for $\leqq, <, \ll$.)

The non-negative orthant of $\mathbb{R}^N$ is the set $\Omega = \left\{x \in \mathbb{R}^N \mid x \geqq 0\right\}$. The letter Ω has throughout this volume the meaning introduced here.

Definitions

Consider a subset S of $\mathbb{R}^N$. To describe those of its points that are infinitely far from 0 one introduces the concept of its asymptotic cone, as follow. Let M be a positive real number, and denote by S^M the set $\{x \in S \mid \|x\| \geqq M\}$ of vectors in S whose norm is greater than M. Let $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing S^M (i.e., the intersection of all the closed cones with vertex 0 containing S^M). The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$, i.e.,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right);$$

it is clearly a closed cone with vertex 0.

Definitions

For a given subset of $\mathbb{R}^N$ we are interested in studying its **behavior at infinity**. This leads to the concepts of asymptotic cone. There are three kinds of definition on asymptotic cone.

Definition 1.1. The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right),$$

where $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing $S^M = \{x \in S \mid \|x\| \geqq M\}$ (i.e., the intersection of all the closed cones with vertex 0 containing S^M).

Definitions

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned}
\Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\
&= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \\
&= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S}^M\right\}
\end{aligned}$$

Definitions

We just take the third definition from Alfred Auslender and Marc Teboulle (2003) *Asymptotic Cones and Functions in Optimization and Variational Inequalities*.

- Let $\{x_k; k \in \mathbb{N}\}$ be a sequence in $\mathbb{R}^N$. We are interested in knowing how to handle situations when the sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is unbounded.
- To derive some convergence properties, we are led to consider directions $d_k = \dfrac{x_k}{\|x_k\|}$ with $x_k \neq 0$ for $k \in \mathbb{N}$.
- From classical analysis, the Bolzano-Weierstrass theorem implies that we can extract a convergent subsequence $d = \lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k$ with $\mathbb{K} \subseteq \mathbb{N}$ and $d \neq 0$.
- Now suppose that the sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is such that $\|x_k\| \to \infty$. Then there exists $t_k = \|x_k\|$, $k \in \mathbb{K} \subseteq \mathbb{N}$, such that $\lim\limits_{k\to\infty} t_k = \infty$ and $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$.

Definitions

A sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is said to converge to a direction $d \in \mathbb{R}^N$ if there exists $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$, with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$.

Definition 1.2. Let C be a non-empty set in $\mathbb{R}^N$. Then the asymptotic cone of the set C, denoted by C_∞, is the set of vectors $d \in \mathbb{R}^N$ that are limits in direction of the sequences $\{x_k; k \in \mathbb{N}\} \subseteq C$, namely

$$C_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{ with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

Definitions

We just take the second definition from R. Tyrrell Rockafellar (1970) *Convex Analysis*.

- The Definition is introduced for convex sets, but the introduction of the asymptotic cone is really intuitive.
- Closed bounded subsets of $\mathbb{R}^N$ are usually easier to work with than unbounded ones. When the sets are convex, however, the difficulties with unboundedness are very much less, and that is fortunate, since so many of the sets we need to consider, like epigraphs, are unbounded by their nature.

Definitions

- Unbounded closed convex sets have a simple behavior "at infinity", according to one's intuition.
- Suppose that C is such a set and x is a point of C. It seems that C must actually contain some entire half-line starting at x, or the unboundedness would be contradicted.
- The directions of such half-lines seem not to depend on x: the half-lines in C starting at a different points are apparently just the translates of those starting at x.
- These directions in which C recedes indefinitely might possibly be thought of as ideal points of C lying at infinity, "horizon points", after the fashion of projective geometry.
- The half-lines in C starting at x could then be interpreted as the segments joining x with such ideal points of C.

Definitions

- Let us first see how the concept of "direction" can be formalized.
- Each closed half-line in $\mathbb{R}^N$ should have a "direction", and two should have the same "direction" if and only if they are translates of each other.
- We therefore define a direction of $\mathbb{R}^N$ simply to be an equivalence class of the collection of all closed half-lines of $\mathbb{R}^N$ under the equivalence relation "half-line L_1 is a translate of half-line L_2".
- The direction of the half-line $\{x + \lambda y : \lambda \geqq 0\}$, where $y \neq 0$, is then by definition the set of all translates of the half-line, and that is independent of x. We shall also call this the direction of y.
- Two vectors in $\mathbb{R}^N$ have the same direction if and only if they are positive scalar multiples of each other.
- The zero vector has no direction. It is clear what one would mean by the opposite of a given direction.

Definitions

Definition 1.3. Let C be a non-empty convex set in $\mathbb{R}^N$.

- We shall say that C recedes in the direction D if C includes all the half-lines in the direction D which start at points of C. In other words, C recedes in the direction of y, where $y \neq 0$, if and only if $x + \lambda y \in C$ for every $\lambda \geqq 0$ and $x \in C$.
- The set of all vectors $y \in \mathbb{R}^N$ satisfying the latter condition, including $y = 0$, will be called the recession cone of C.
- The **recession cone** of C will be denoted by $\mathbf{0}^+ C$.

$$\mathbf{0}^+ C = \left\{ y \in \mathbb{R}^N : \text{for all } x \in C, x + \lambda y \in C \text{ for all } \lambda \geqq 0 \right\}$$

- Directions in which C recedes will also be referred to as directions of recession of C.
- The recession cone of $\overline{C}$ has elsewhere been called the asymptotic cone of C.

$$\mathbf{A}C = \mathbf{0}^+ \overline{C}.$$

Proposition of Definition 2

Proposition 2.1. Let $C \subseteq \mathbb{R}^N$ be non-empty.

(1) C_∞ is a closed cone.

(2) $\left(\overline{C}\right)_\infty = C_\infty$.

(3) If C is a cone, then $C_\infty = \overline{C}$.

Proposition of Definition 2

Proof of (1). $0 \in C_\infty$ because if we fix $x_k = x \in C$, for any $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$, we have $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$.

If $d \in C_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. For any $\lambda > 0$, we have $\left\{\frac{1}{\lambda}t_k; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\frac{1}{\lambda}t_k \to \infty$, we have $\lim_{k\to\infty} \frac{x_k}{\frac{1}{\lambda}t_k} = \lambda d$, i.e. $\lambda d \in C_\infty$.

Hence C_∞ is a cone.

Proposition of Definition 2

Next, we need to show C_∞ is closed.

For any $d \in \overline{C_\infty}$, there exists $\{d^q; q \in \mathbb{N}\} \subseteq C_\infty$ such that $d^q \to d$. For any d^q, we know there exist $\left\{x_k^q; k \in \mathbb{N}\right\} \subseteq C$ and $\left\{t_k^q; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\lim\limits_{k\to\infty} t_k^q = \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k^q}{t_k^q} = d^q$.

Then there exists $K_q \in \mathbb{N}$ such that for all $k \geqq K_q$, we have $t_k^q > q$ and $\left\| \dfrac{x_k^q}{t_k^q} - d^q \right\| < \dfrac{1}{q}$. Let $\left\{x_q = x_{K_q}^q\right\} \subseteq C$ and $\left\{t'_q = t_{K_q}^q\right\} \subseteq \mathbb{R}$, we have $t'_q = t_{K_q}^q > q$ hence $\lim\limits_{q\to\infty} t'_q = \infty$.

$$\left\| \frac{x_q}{t'_q} - d \right\| \leqq \left\| \frac{x_q}{t'_q} - d^q \right\| + \|d^q - d\| < \frac{1}{q} + \|d^q - d\| \to 0$$

hence $\lim\limits_{q\to\infty} \dfrac{x_q}{t'_q} = d$, which implies $d \in C_\infty$. Hence C_∞ is a closed cone.

Proposition of Definition 2

Proof of (2). It's trivial that $C_\infty \subseteq \left(\overline{C}\right)_\infty$. Conversely, let $d \in \left(\overline{C}\right)_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \overline{C}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. For any $x_k \in \overline{C}$, there exists $x'_k \in C$ such that $\|x'_k - x_k\| < \frac{1}{k}$. Hence

$$\begin{aligned}
\lim_{k\to\infty} \left\| \frac{x'_k}{t_k} - d \right\| &\leqq \lim_{k\to\infty} \left[\left\| \frac{x'_k}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \right] \\
&< \lim_{k\to\infty} \frac{1}{k\,|t_k|} + \lim_{k\to\infty} \left\| \frac{x_k}{t_k} - d \right\| \longrightarrow 0
\end{aligned}$$

hence $\lim_{k\to\infty} \frac{x'_k}{t_k} = d$, i.e. $d \in C_\infty$. Hence $\left(\overline{C}\right)_\infty = C_\infty$.

Proposition of Definition 2

Proof of (3). We just need to prove that if C is a closed cone, then $C_\infty = C$. Then for any cone C, we have $C_\infty = \left(\overline{C}\right)_\infty = \overline{C}$ as $\overline{C}$ is also a cone.

If C is a closed cone, then for any $x \in C$, for any $k > 0$, we have $kx \in C$. Let $x_k = kx$ and $t_k = k$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = x$, hence $x \in C_\infty$, i.e. $C \subseteq C_\infty$.

For any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Note that C is a cone, $\dfrac{x_k}{t_k} \in C$, then $d \in \overline{C} = C$. Hence $C_\infty = C$. $\square$

Proposition of Definition 2

Theorem 2.1. If S is a subset of $\mathbb{R}^N$, then

$$\mathbf{A}S = S_\infty$$

where

$$\mathbf{A}S = \bigcap_{k \geqq 0} \Gamma\left(S_M\right)$$

with $S^M = \{x \in S \mid \|x\| \geqq M\}$ and
$\Gamma\left(S^M\right) = \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}}$ and

$$S_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}.$$

Proposition of Definition 2

Proof. For any $d \in S_\infty$. If $d = 0$, then $d \in \mathbf{A}S$ is trivial. If $d \neq 0$, then there exist $\{x_n\} \subseteq S$ and $\{t_n\} \subseteq \mathbb{R}$ with $t_n \to \infty$ such that $\lim_{n\to\infty} \frac{x_n}{t_n} = d$. There exists $N \in \mathbb{N}$ such that for all $n \geqq N$, we have $\left\| \frac{x_n}{t_n} \right\| > \frac{\|d\|}{2} > 0$. Then for any $M \geqq 0$, as $t_n \to \infty$, there exists $N_M > 0$ such that for all $n \geqq N_M$, we have $\|x_n\| \geqq M$. Hence $x_n \in S^M$. Hence $\frac{x_n}{t_n} \in \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}$ for all $n \geqq N_M$. As $\lim_{n\to\infty} \frac{x_n}{t_n} = d$, we have $d \in \Gamma\left(S^M\right)$ for any $M \geqq 0$. Thus,

$$S_\infty \subseteq \bigcap_{M\geqq 0} \Gamma\left(S^M\right) = \mathbf{A}S$$

Conversely, if $d \in \mathbf{A}S$ with $d \neq 0$, then for all $M \geqq 0$, $d \in \Gamma\left(S^M\right) = \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}}$, so there exists $\alpha_M x_M$ where $\alpha_M > 0$ and $x_M \in S^M$ such that $\|d - \alpha_M x_M\| < \frac{1}{M}$. Then let $t_M = \frac{1}{\alpha_M}$, if we can prove $t_M \to \infty$, i.e. $\alpha_M \to 0$, then we are done. As $\|\alpha_M x_M\| \leqq \|d\| + \|d - \alpha_M x_M\|$, then

$$|\alpha_M| \leqq \frac{\|d\| + \|d - \alpha_M x_M\|}{\|x_M\|} < \frac{\|d\| + \frac{1}{M}}{M} \to 0.$$

We are done! $\square$

Proposition of Definition 2

Proposition 2.2. Let $C \subseteq \mathbb{R}^N$ be non-empty and define the normalized set

$$C_N = \left\{ d \in \mathbb{R}^N \mid \text{there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ with } \|x_k\| \to \infty \text{ such that } d = \lim_{k\to\infty} \frac{x_k}{\|x_k\|} \right\}$$

then $C_\infty = \text{Cone}\,(C_N)$, where for any set C, $\text{Cone}\,(C_N) = \left\{ \lambda x \in \mathbb{R}^N \mid x \in C_N, \lambda \geqq 0 \right\}$.

Proof. Clearly, one always has $\text{Cone}\,(C_N) \subseteq C_\infty$ as $C_N \subseteq C_\infty$.

Conversely, let $0 \neq d \in C_\infty$. Then there exist $x_k \in C$ and $t_k \to \infty$, such that

$$d = \lim_{k\to\infty} \frac{x_k}{t_k} = \lim_{k\to\infty} \frac{\|x_k\|}{t_k} \frac{x_k}{\|x_k\|}$$

with $\|x_k\| \to \infty$. Thus the sequence $\left\{ \frac{\|x_k\|}{t_k}; k \in \mathbb{N} \right\}$ is a positive bounded sequence, and by the Bolzano–Weierstrass theorem, there exists a subsequence $\left\{ \frac{\|x_k\|}{t_k}; k \in \mathbb{K} \right\}$ with $\mathbb{K} \subseteq \mathbb{N}$ such that

$$\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{\|x_k\|}{t_k} = \lambda \geqq 0.$$

we also have

$$d = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{t_k} = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{\|x_k\|}{t_k} \frac{x_k}{\|x_k\|},$$

which means that $d = \lambda d_N$ with $d_N = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} \in C_N$, namely $d \in \text{Cone}\,(C_N)$. □

Proposition of Definition 2

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \left\{d \in \mathbb{R}^N : \|d\| = 1\right\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! $\square$

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S + \{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S + \{x\})$, we just need to let $S = (S + \{x\}) + \{-x\}$ and we arrive at the conclusion. Let $T = S + \{x\}$.

For any $d \in \mathbf{A}(S + \{x\})$, $d \in (S + \{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S + \{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\| \frac{x'_k}{t_k} - d \right\| \leqq \left\| \frac{x'_k}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| = \left\| \frac{x}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S + \{x\}) \subseteq \mathbf{A}S$. $\square$

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

Asymptotic Cone

Theorem 3.2. If $E \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}(S + E)$.

Proof. Let $x \in E$. Then $\mathbf{A}S = \mathbf{A}(S + \{x\}) \subseteq \mathbf{A}(S + E)$. □

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. $\square$

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

Proof. $y \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$, then, for any $M \geqq 0$, $y \in \Gamma\left(\left(\prod_{n=1}^{N} S_n\right)^M\right)$, then

$$
\begin{aligned}
y \quad &\in \quad \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \left(\prod_{n=1}^{N} S_n\right)^M\right\}} = \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{\prod_{n=1}^{N} S_n^M}\right\} \\
&\subseteq \quad \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \prod_{n=1}^{N} \overline{S_n^M}\right\} = \prod_{n=1}^{N} \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S_n^M}\right\} \\
&= \quad \prod_{n=1}^{N} \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S_n^M\right\}} = \prod_{n=1}^{N} \Gamma\left(S_n^M\right)
\end{aligned}
$$

hence $y \in \bigcap_{M \geqq 0}\left[\prod_{n=1}^{N} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N}\left[\bigcap_{M \geqq 0} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N} \mathbf{A}\left(S_n\right)$. Therefore $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

□

Asymptotic Cone

Theorem 3.4. Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. $\square$

Asymptotic Cone

Theorem 3.4. For all $j \in \mathcal{J}$, $S_j \subseteq \mathbb{R}^N$, if $\bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$, then $\bigcap_{j\in\mathcal{J}} S_j$ is bounded.

Proof. For all $j \in \mathcal{J}$, $\bigcap_{j\in\mathcal{J}} S_j \subseteq S_j$, then $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \mathbf{A}(S_j)$. Whence $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap_{j\in\mathcal{J}} S_j\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap_{j\in\mathcal{J}} S_j$ is bounded. $\square$

Asymptotic Cone

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, a sufficient condition for the set $L(C)$ to be closed is that $L^{-1}\{0\} \cap C_\infty = \{0\}$. Under this condition one has $L(C_\infty) = L(C)_\infty$, although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, if $L^{-1}\{0\} \cap C_\infty = \{0\}$, then

(1) the set $L(C)$ is closed;

(2) $L(C_\infty) = L(C)_\infty$.

Although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Asymptotic Cone

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{N} \right\}$ is bounded, so we can extract a subsequence of $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{K} \right\}$ such that $\lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

Asymptotic Cone

Assuming henceforth that the condition holds, we prove now that $L(C)$ is a closed set. Suppose $y_k \to y$ with $y_k \in L(C)$, or in other words, $y_k = L(x_k)$ for some $x_k \in C$. The sequence $\{L(x_k)\}$ is bounded, so the sequence $\{x_k; k \in \mathbb{N}\}$ must be bounded as well, for the reasons just given, and it therefore has a cluster point x. We have $x \in C$ (because C is closed) and $y = L(x)$ (again because linear mappings are continuous), and therefore $y \in L(C)$.

Asymptotic Cone

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

Asymptotic Cone

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. $\square$

Asymptotic Cone

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, a sufficient condition for $\sum_{n=1}^{N} C_n$ to be closed is the non-existence of vectors $x_n \in (C_n)_\infty$ such that $\sum_{n=1}^{N} x_n = 0$, except with $x_n = 0$ for all n. Then also

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty .$$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, if $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

(1) $\sum_{n=1}^{N} C_n$ is closed;

(2) $\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$.

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

Asymptotic Cone

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty$$

$$L((x_1, \cdots, x_N)) = 0 \quad \text{and} \quad x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N$$

$$x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0$$

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then $x_n = 0$ for $n = 1, \cdots, N$, and hence

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

Asymptotic Cone

Theorem 3.6. $S_1, \cdots, S_j, \cdots, S_J$ are closed subsets of $\mathbb{R}^N$, if their asymptotic cones $\mathbf{A}(S_1), \cdots, \mathbf{A}(S_j), \cdots, \mathbf{A}(S_J)$ are positively semi-independent, then their sum $\sum_{j=1}^{J} S_j$ is closed.

Proof. We first verify the setting of $J = 2$. $\mathbf{A}(S_1)$ and $\mathbf{A}(S_2)$ are positively semi-independent if and only if $\mathbf{A}(S_1) \cap \mathbf{A}(-S_2) = \{0\}$.

Suppose $\mathbf{A}(S_1) \neq \{0\} \neq \mathbf{A}(S_2)$, then S_1 and S_2 are unbounded. We assume that $\{z^K; K \in \mathbb{R}_+\} \subseteq S_1 + S_2$ and $z^K \to z^0$. According to the definition, for any $K \in \mathbb{R}_+$, there exist $x_1^K \in S_1$ and $x_2^K \in S_2$ such that $z^K = x_1^K + x_2^K$.

Asymptotic Cone

If $\{x_1^K; K \in \mathbb{R}_+\}$ is unbounded in S_1, then for any $K \in \mathbb{R}_+$, $x_1^K \in S_1^K$, $x_1^K \in \Gamma(S_1^K)$, $\|x_1^K\| \to \infty$, then $x_2^K = z^K - x_1^K$ and $\|x_2^K\| \to \infty$. For any $K \in \mathbb{R}_+$, $-x_2^K \in -S_2^K$, $-x_2^K \in \Gamma(-S_2^K)$, and

$$\frac{x_1^K}{\|x_1^K\|} = \frac{z^K}{\|x_1^K\|} - \frac{x_2^K}{\|x_1^K\|}, \quad \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \quad \text{and} \quad -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

then

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} \frac{z^K}{\|x_1^K\|} + \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|},$$

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \qquad \text{and} \qquad \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

that is,

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} = e \neq 0, \quad e \in \Gamma(S_1^K) \quad \text{and} \quad e \in \Gamma(-S_2^K)$$

Hence $0 \neq e \in \Gamma(S_1^K) \cap \Gamma(-S_2^K) = \{0\}$, contradiction!

Asymptotic Cone

Therefore $\{x_1^K; K \in \mathbb{R}_+\}$ id bounded in S_1, then there exists a convergent subsequence $\{x_1^{K_n}; K_n \in \mathbb{R}_+\} \subseteq \{x_1^K; K \in \mathbb{R}_+\}$ such that $x_1^{K_n} \to x_1^0 \in S_1$. On the other hand, for any $K_n \in \mathbb{R}_+$, $x_2^{K_n} = z^{K_n} - x_1^{K_n}$, then $x_2^{K_n} \to x_2^0 \in S_2$, and hence $z^{K_n} \to x_1^0 + x_2^0 \in S_1 + S_2$. Thus $z^0 = x_1^0 + x_2^0 \in S_1 + S_2$, and hence $S_1 + S_2$ is closed.

The general conclusion holds from mathematical induction. $\square$

Proposition of Definition 3

Associated with the asymptotic cone C_∞ is the following related concept, which will help us in simplifying the definition of C_∞ in the particular case where $C \subseteq \mathbb{R}^N$ is assumed convex.

Definition Let $C \subseteq \mathbb{R}^N$ be non-empty and define

$$C_\infty^1 = \left\{ d \in \mathbb{R}^N : \text{for all } t_k \to \infty \text{ there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \right.$$
$$\left. \text{such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

We say that C is asymptotically regular if $C_\infty = C_\infty^1$.

Proposition of Definition 3

Proposition 4.1. Let C be a non-empty convex set in $\mathbb{R}^N$. Then C is asymptotically regular.

Proposition of Definition 3

Proof. The inclusion $C_\infty^1 \subseteq C_\infty$ clearly holds from the definitions of C_∞^1 and C_∞, respectively.

Let $d \in C_\infty$. Then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t'_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t'_k \to \infty$ such that $d = \lim_{k\to\infty} \frac{x_k}{t'_k}$. Let $x \in C$ and define

$$d_k = \frac{1}{t'_k}(x_k - x),$$

then we have

$$d = \lim_{k\to\infty} d_k$$

and $x + t'_k d_k = x_k \in C$.

Proposition of Definition 3

Now let $\{t_k; k \in \mathbb{N}\}$ be an arbitrary sequence such that $\lim_{k\to\infty} t_k = \infty$. For any fixed $N \in \mathbb{N}$, there exists $k(N)$ with $\lim_{N\to\infty} k(N) = \infty$ such that $t_N \leqq t'_{k(N)}$, and since C is convex, we have

$$z_N = x + t_N d_{k(N)} \in C.$$

As z_N is on the closed segment $\left[x, x + t'_{k(N)} d_{k(N)}\right]$. Hence,

$$d = \lim_{N\to\infty} \frac{z_N}{t_N},$$

showing that $d \in C^1_\infty$. $\square$

Proposition of Definition 3

Proposition 4.2. Let C be a non-empty convex set in $\mathbb{R}^N$. Then the asymptotic cone C_∞ is a closed convex cone.

Proof. By Proposition 2.1, we have shown that C_∞ is a closed cone. We just need to verify C_∞ is convex.

By Proposition 4.1, we know that for any d_1 and d_2 in C_∞^1, for any $t_k \to \infty$, there exists x_k^1 and x_k^2 in C such that

$$\lim_{k\to\infty} \frac{x_k^1}{t_k} = d_1 \qquad \text{and} \qquad \lim_{k\to\infty} \frac{x_k^2}{t_k} = d_2$$

Then for any $\lambda \in [0,1]$, we have $\lambda x_k^1 + (1-\lambda)\, x_k^2 \in C$ and

$$\lim_{k\to\infty} \frac{\lambda x_k^1 + (1-\lambda)\, x_k^2}{t_k} = \lambda d_1 + (1-\lambda)\, d_2$$

hence $\lambda d_1 + (1-\lambda)\, d_2 \in C_\infty^1$. Hence C_∞^1 is convex, hence C_∞ is convex. $\square$

Proposition of Definition 3

Proposition 4.3. If C is a non-empty subset of $\mathbb{R}^N$, then

$$\mathbf{0}^+ C \subseteq C_\infty$$

where

$$\mathbf{0}^+ C = \left\{ y \in \mathbb{R}^N : \text{for all } x \in C, x + \lambda y \in C \text{ for all } \lambda \geqq 0 \right\}$$

and

$$C_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \right.$$
$$\left. \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

If C is non-empty closed and convex, then

$$\mathbf{0}^+ C = C_\infty.$$

So if C is non-empty convex, then

$$\mathbf{0}^+ \overline{C} = C_\infty.$$

Proposition of Definition 3

Proof. For any $y \in \mathbf{0}^+ C$, for any $k > 0$, let $t_k = k \to \infty$ and $x_k = x + ky$ for any $x \in C$. Then

$$\left\| \frac{x_k}{t_k} - y \right\| = \left\| \frac{x}{k} \right\| \to 0$$

hence $\lim_{k\to\infty} \frac{x_k}{t_k} = y$, which implies that $\mathbf{0}^+ C \subseteq C_\infty$.

If C is non-empty, closed and convex, then for any $d \in C_\infty$, we need to show that for all $x \in C$ and $t > 0$, we have $x + td \in C$. As there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let $d_k = \frac{x_k - x}{t_k}$, then we have $x_k = x + t_k d_k \in C$ and

$$\lim_{k\to\infty} \frac{x_k}{t_k} = \lim_{k\to\infty} \frac{x + t_k d_k}{t_k} = \lim_{k\to\infty} \left(d_k + \frac{x}{t_k} \right) = d$$

hence

$$\lim_{k\to\infty} d_k = d.$$

Proposition of Definition 3

Take k sufficiently large such that $t \leqq t_k$, then since C is convex, we have

$$x + td_k = \left(1 - \frac{t}{t_k}\right) x + \frac{t}{t_k} (x + t_k d_k) \in C.$$

Passing to the limit, we thus have $x + td = \lim_{k\to\infty} (x + td_k) \in C$ as C is closed. So $d \in \mathbf{0}^+ C$. Hence $\mathbf{0}^+ C = C_\infty$.

So if C is non-empty convex, we have $\overline{C}$ is non-empty, closed and convex

$$\mathbf{0}^+ \overline{C} = \left(\overline{C}\right)_\infty = C_\infty$$

by Proposition 2.1. $\square$

Proposition of Definition 3

If C is not closed, the theorem may fail. Consider

$$C = \{(x_1, x_2) : 0 \leqq x_1 < 1, x_2 \geqq 0\} \cup \{(1, 0)\}$$

By using the definition of direction of recession, it can be verified that C has no nonzero directions of recession, i.e.

$$\mathbf{0}^+ C = \varnothing$$

but we can see that

$$C_\infty = \mathbf{0}^+ \overline{C} = \{(0, x) : x \in \mathbb{R}\}.$$

Proposition of Definition 3

Proposition 4.4. For any non-empty closed convex set $C \subseteq \mathbb{R}^N$, one has $C = C + C_\infty$.

Proof. The inclusion $C \subseteq C + C_\infty$ is clear because $0 \in C_\infty$.

Let $x \in C + C_\infty$. Then there exists $c \in C$, $d \in C_\infty$ with $x = c + d$. Thus, there exists $d_k \in C$, $t_k \to \infty$ such that $\dfrac{d_k}{t_k} \to d$, which implies that for k sufficiently large

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \in C$$

and

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \to c + d = x \in C$$

as C is closed. $\square$

Proposition of Definition 3

Corollary A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. As $0 \in C$ and $\{0\} + C_\infty = C_\infty$. As $C = C + C_\infty$, we have

$$C_\infty \subseteq C.$$

□

Proposition of Definition 3

Proposition 4.5. A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\dfrac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda \dfrac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. $\square$

Proposition of Definition 3

Proposition 4.6. Let $S \subseteq \mathbb{R}^N$ be a closed convex set. For $x_0 \in S$, define the cone C as follows:

$$C = \max\{C' \mid x_0 + C' \subseteq S\},$$

then C is an asymptotic cone, that is

(1) $C = \bigcap_{\lambda>0} \lambda(S - x_0)$;

(2) C is independent of x_0;

(3) $C = \mathbf{A}S$.

Proposition of Definition 3

Proof of (1). $C \subseteq S - x_0 = S'$ and $0 \in S'$.

$$
\begin{aligned}
y \in C &\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\implies \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\implies y \in \bigcap_{\lambda>0} \lambda S'
\end{aligned}
$$

then $C \subseteq \bigcap_{\lambda>0} \lambda(S - x_0)$. Conversely,

$$
\begin{aligned}
y \in \bigcap_{\lambda>0} \lambda S' &\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\implies \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\implies y \in C
\end{aligned}
$$

then $\bigcap_{\lambda>0} \lambda(S - x_0) \subseteq C$. $\square$

Proposition of Definition 3

Proof of (2). Suppose $x_0 \in S$, $y \in S \Longrightarrow \mathbb{R}_+ y \subseteq C \subseteq S'$. For any $x \in S$,

$$x_0 + \mathbb{R}_+ y \subseteq S \Longleftrightarrow x + \mathbb{R}_+ y \subseteq S$$

then C is independent of x_0.

Proof of (3).

$$\mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S' \Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \Longrightarrow \mathbb{R}_+ y \in \mathbf{A}S' = \mathbf{A}S$$

then $\bigcap_{\lambda>0} \lambda S' \subseteq \mathbf{A}S$. Conversely,

$$\begin{aligned}
\mathbb{R}_+ y \in \mathbf{A}S = \mathbf{A}S' &\Longrightarrow \text{for any } M > 0, \mathbb{R}_+ y \subseteq \Gamma(S'^M) \\
&\Longrightarrow \text{for any } M > 0, [\alpha_M, \infty) y \subseteq S'^M \\
&\Longrightarrow \mathbb{R}_+ y \subseteq S' \\
&\Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \\
&\Longrightarrow \mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S'
\end{aligned}$$

then $\mathbf{A}S \subseteq \bigcap_{\lambda>0} \lambda S'$. $\square$